Building a Multi-modal Spatiotemporal Expert for Zero-shot Action Recognition with CLIP

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Abstract

Zero-shot action recognition (ZSAR) demands models that align visual dynamics with rich action semantics, yet existing CLIP-based methods struggle on two fronts: CLIP's spatial attention processes frames independently, missing cross-frame temporal dynamics, and coarse category-label text provides insufficient semantic discrimination for novel actions. We present **STDD** (**S**patio**T**emporal **D**ynamic **D**uo), a method that converts CLIP into a multi-modal spatiotemporal expert through two complementary modules. On the vision side, a parameter-free *Space-Time Cross Attention* (STCA) augments each ViT block with cross-frame dynamics via Window Shift Masking, Multi-scale Channel Mixing, spatial padding, and residual shortcutting—preserving the original $\mathcal{O}(TN^2)$ complexity. On the text side, an *Action Semantic Knowledge Graph* (ASKG) built with large language models decomposes action labels into structured spatial and temporal prompts encoding object relations and sub-action sequences. A fine-grained frame-prompt symmetric alignment objective, coupled with CLIP feature distillation, ensures robust zero-shot transfer. Trained on Kinetics-400, STDD achieves state-of-the-art performance on UCF101 (93.4%), HMDB51 (68.7%), and Kinetics-600 (75.1%), demonstrating consistent gains across all standard ZSAR protocols.

1 Introduction

Action recognition in unconstrained videos is a cornerstone of visual understanding, with broad applications in surveillance, sports analysis, and human-computer interaction. The *zero-shot* variant—recognizing actions from categories entirely unseen during training—is particularly compelling, as it mirrors the human ability to identify new activities from conceptual descriptions alone. With the advent of large-scale vision-language pre-training, CLIP (Radford et al. 2021) has emerged as a powerful foundation for zero-shot transfer, aligning image and text representations through contrastive learning on hundreds of millions of web image-text pairs.

Despite CLIP's remarkable spatial alignment, directly applying it to ZSAR reveals two critical limitations, illustrated in Figure 1. **First**, CLIP's Vision Transformer (ViT) (Dosovitskiy et al. 2020) backbone processes each frame independently: spatial self-attention operates within a single frame, so cross-frame temporal dynamics—motion trajectories, sequential sub-actions, object state changes—are entirely invisible to the model. Temporal pooling over per-frame features provides only a coarse, order-agnostic video representation, insufficient for distinguishing fine-grained actions that share similar static appearances (*e.g.*, , "archery" *vs.* "throwing a disc"). **Second**, standard CLIP-based ZSAR approaches encode action labels as simple category-text prompts (*e.g.*, , "a video of archery"), which lack the semantic richness needed to discriminate novel actions. Actions involve complex interaction of objects, scene contexts, and temporal sequences of sub-actions, none of which are captured by a single label string. This semantic impoverishment limits generalization to unseen categories.

To address both challenges simultaneously, we propose the **Spatiotemporal Dynamic Duo (STDD)**, which extends CLIP from a spatial expert to a full multi-modal spatiotemporal expert. Our core insight is that temporal dynamics and semantic augmentation are synergistic: richer text semantics provide better alignment targets, while improved temporal features make those targets more reachable.

On the vision side, we introduce *Space-Time Cross Attention* (STCA), a parameter-free module inserted into every ViT block. STCA operates in four steps: *Window Shift Masking* (WSM) sparsifies and interleaves spatial tokens across frames to enable cross-frame token interactions; *Multi-scale Channel Mixing* (MCM) propagates temporal cues at multiple temporal scales by shifting channel partitions between adjacent frames; *spatial padding* restores the full token layout after sparse attention; and a *residual shortcut* fuses the temporal dynamics back into the original spatial features. Crucially, STCA introduces *no additional parameters* and maintains the same $\mathcal{O}(TN^2)$ complexity as single-frame attention.

On the text side, we construct an *Action Semantic Knowledge Graph* (ASKG) using large language models (Achiam et al. 2023). For each action class, the ASKG decomposes the action into static concepts (objects, appearances, scene relations) and dynamic concepts (sub-actions, motion sequences), generating two complementary prompt sets: *spatial prompts* describing object-object and action-object relations, and *temporal prompts* encoding action-action sequential relations. This spatiotemporal text augmentation provides the semantic precision needed to separate fine-grained

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novel actions.

Training aligns frame-level video features with these structured text prompts through a symmetric max-over-frames/prompts scoring mechanism, while a CLIP-based feature distillation loss preserves the strong zero-shot generalizability of the original model.

Our main contributions are:

- A **parameter-free STCA** that transforms CLIP's single-frame spatial attention into space-time cross-attention with no additional parameters or computational overhead.
- An **ASKG-driven spatiotemporal text augmentation** that decomposes action semantics into complementary spatial and temporal prompt sets using LLM-generated knowledge graphs.
- A **fine-grained frame-prompt symmetric alignment** with CLIP feature distillation for robust zero-shot transfer.
- **State-of-the-art results** across all standard ZSAR protocols on UCF101, HMDB51, and Kinetics-600, surpassing prior CLIP-based methods by up to $+1.8\%$.

2 Related Work

CLIP-based Video Understanding. The success of CLIP (Radford et al. 2021) on image-text matching has sparked significant efforts to adapt it for video. ActionCLIP (Wang, Xing, and Liu 2021) reformulates action recognition as a new-paradigm matching problem between videos and text. X-CLIP (Ni et al. 2022) extends CLIP with cross-frame attention and multi-frame integration. ViFi-CLIP (Rasheed et al. 2023) demonstrates that simple fine-tuning of CLIP is a surprisingly strong video baseline. Vita-CLIP (Wasim et al. 2023) and ViLT-CLIP (Wang et al. 2024) explore multimodal prompt learning to balance supervised and zero-shot performance. More recent approaches such as Open-VCLIP (Weng et al. 2023) and Open-VCLIP++ (Wu et al. 2024) improve temporal modeling through continual fine-tuning strategies. Our STCA differs from these methods by inserting parameter-free cross-frame token interactions directly into each ViT block, requiring no additional parameters.

Temporal Modeling in Video Transformers. Adapting image transformers to video requires injecting temporal reasoning without sacrificing spatial representations. TimeSformer (Bertasius, Wang, and Torresani 2021) proposes divided space-time attention, separating temporal and spatial attention heads across alternating transformer layers, while Video Swin Transformer (Liu et al. 2022) extends shifted window attention to 3D spatiotemporal windows for hierarchical feature capture. Lightweight approaches such as EVL (Lin et al. 2022) and ST-Adapter (Pan et al. 2022) inject temporal reasoning into frozen image ViTs via compact adapter modules, preserving zero-shot capabilities while adding limited parameters. AIM (Yang et al. 2023) further introduces attention-based temporal modules with spatial pooling. Unlike these approaches, STCA achieves cross-frame interaction entirely by reorganizing existing computations, adding zero parameters while still capturing multi-scale temporal dynamics.

Zero-Shot Action Recognition. Early ZSAR methods rely on hand-crafted semantic embeddings (Wang and Chen 2017) or generative approaches (Mishra et al. 2018) with uni-modal visual features. With the rise of VLMs, CLIP-based methods have become dominant. ER-ZSAR (Chen and Huang 2021) uses elaborative descriptions for richer semantics. BIKE (Wu et al. 2023) explores bidirectional cross-modal knowledge via attribute association. FROSTER (Huang et al. 2024) treats frozen CLIP as a regularization teacher to prevent over-fitting to seen categories. OST (Chen et al. 2024) refines text with optimal spatiotemporal descriptors, and AP-CLIP (Jia et al. 2024) generates action-conditioned prompts with LLMs. MAXI (Lin et al. 2023) employs visual captions for unsupervised finetuning. VicTR (Kahatapitiya et al. 2024) argues for video-conditioned text representations. Our STDD uniquely combines parameter-free temporal dynamics with ASKG-structured prompts, yielding superior ZSAR generalization.

Prompt Learning and Semantic Augmentation. CoOp (Zhou et al. 2022) and its variants learn context vectors for CLIP adaptation. Text4Vis (Wu, Sun, and Ouyang 2023) revisits text-based semantic augmentation for video recognition, showing that spatiotemporal descriptions consistently outperform simple category text. Knowledge graph-based semantic augmentation has been explored in earlier ZSAR works (Gao, Zhang, and Xu 2019; Chen and Huang 2021), where attribute graphs enrich class representations. Our ASKG extends this idea by leveraging LLMs (Achiam et al. 2023) to produce structured spatial and temporal relational prompts, providing more nuanced action semantics than flat attribute lists.

3 Method: Spatiotemporal Dynamic Duo

Figure 2 presents the overall STDD framework, which processes a video clip through two parallel streams. The vision stream encodes frames using a CLIP ViT backbone augmented with STCA modules (Figure 3), while the text stream encodes ASKG-generated spatiotemporal prompts. During training, frame-level visual features are aligned with prompt-level text features via a symmetric scoring objective, with an auxiliary distillation loss from a frozen CLIP teacher.

Space-Time Cross Attention (STCA)

Standard CLIP ViT computes self-attention within each frame independently. Given T video frames, each tokenized into N spatial tokens plus one `[CLS]` token, the per-frame self-attention operates on sequences of length $N+1$, with complexity $\mathcal{O}(TN^2)$. While this processes all frames in parallel efficiently, it cannot model any cross-frame interaction.

STCA augments each ViT residual attention block to enable cross-frame token interaction without introducing any learnable parameters. Let $\mathbf{x} \in \mathbb{R}^{(N+1)\times BT\times D}$ be the token sequence for a batch of B clips with T frames each.

Step 1: Window Shift Masking (WSM). We partition the spatial tokens of each frame into non-overlapping spatial windows of stride $s \times s$. For each window position, we select a subset of tokens according to a cyclic shift pat-

tern that varies across frames—*i.e.*, , tokens from different frames are selected from complementary window cells. This sparse interleaving reduces the token count from N to $N/2$ per frame while ensuring that the retained tokens of adjacent frames cover complementary spatial locations. Formally, for a mask stride of $s_1 \times s_2$, we construct a binary mask $\mathbf{m} \in \{0,1\}^{B\times T\times N}$ such that each spatial cell is covered by exactly one frame in any local temporal window. The masked token sequence $\mathbf{x}_{\text{mask}}$ thus interleaves spatial contexts across frames.

Step 2: Multi-scale Channel Mixing (MCM). Before computing attention on the masked tokens, we inject explicit temporal cues by shifting channel partitions between adjacent frames. Specifically, we partition the D-dimensional token features into three channel segments: the first D/f channels are shifted one step forward in time (*i.e.*, , frame t receives channel features from frame $t-1$), the next D/f channels are shifted one step backward, and the remaining channels are left unchanged. For multi-scale mixing at scale s, we perform s successive such shifts with progressively halved channel widths $(D/f, D/2f, \ldots)$, allowing short- and medium-range temporal cues to coexist. The mixed tokens $\mathbf{x}_{\text{mix}}$ are then passed through the standard self-attention layer.

Step 3: Spatial Padding. After attention on the mixed masked tokens, we restore the full $N+1$ token layout by filling unselected positions with the output features of the original (unmasked) attention path. This ensures that the subsequent MLP sub-layer operates on a complete spatial token grid, preserving the spatial representation quality.

Step 4: Residual Shortcut. The spatiotemporally enriched, padded features are fused back into the main residual stream. The full forward pass for each STCA block is:

$$\mathbf{x}' = \mathbf{x} + \text{Attn}(\text{LN}(\mathbf{x})), \quad \mathbf{x}_{\text{pad}} = \text{Pad}(\mathbf{x}', \mathbf{x}_{\text{avg}}, \mathbf{m}), \tag{1}$$

$$\mathbf{x}_{\text{out}} = \mathbf{x}_{\text{pad}} + \text{MLP}(\text{LN}(\mathbf{x}_{\text{pad}})), \tag{2}$$

where $\mathbf{x}_{\text{avg}}$ is the average of multi-scale channel-mixed attention outputs. This design introduces *zero new parameters*—no additional weight matrices are added—and the complexity remains $\mathscr{O}(TN^2)$, identical to the original per-frame attention.

Action Semantic Knowledge Graph (ASKG)

Standard CLIP ZSAR encodes action labels as simple category text prompts such as "a video of [action]", which provides insufficient semantic detail for distinguishing fine-grained or compositional actions. We instead construct an *Action Semantic Knowledge Graph* (ASKG) for each action class using an LLM (Achiam et al. 2023).

For each action a, the ASKG encodes three types of relational triples: (i) *object-object relations* capturing static scene structure (*e.g.*, , ⟨beer, poured into, glass⟩); (ii) *action-object relations* describing what entities the action involves (*e.g.*, , ⟨drinking beer, uses, glass⟩); and (iii) *action-action relations* encoding the temporal sequence of sub-actions (*e.g.*, , ⟨opening bottle, precedes, pouring beer⟩). The LLM is prompted with the action label and these triple types to generate a structured graph for each class.

From the ASKG, we derive two complementary sets of natural-language prompts:

- **Spatial prompts** derived from object-object and action-object triples, describing the static spatial context of the action (*e.g.*, , "a video of drinking beer, where beer is poured into a glass").
- **Temporal prompts** derived from action-action sequential triples, describing the dynamic progression of sub-actions (*e.g.*, , "a video of drinking beer, starting with opening a bottle").

Each action class is thus represented by a set $\mathscr{P}_a = \{p_1, \ldots, p_K\}$ of K structured prompts, which are encoded by CLIP's text encoder to produce a prompt feature matrix $\mathbf{T}_a \in \mathbb{R}^{K\times D}$.

Training Objectives

Frame-Prompt Symmetric Alignment. For a video clip with T frames, the STCA-augmented ViT produces frame-level features $\{\mathbf{v}_t\}_{t=1}^{T}$, where $\mathbf{v}_t \in \mathbb{R}^D$. Rather than averaging frame features and matching against a single class text, we use a *symmetric max-over-frames/prompts* scoring function. The similarity between a video and action class a is:

$$s(v,a) = \max_t \max_k \mathbf{v}_t^\top \mathbf{T}_a^{(k)} / \tau, \tag{3}$$

where τ is CLIP's learned temperature. This fine-grained alignment allows each frame to be matched against its most semantically relevant prompt, and vice versa, capturing the compositional nature of actions.

CLIP Feature Distillation. To prevent over-fitting to the Kinetics-400 training distribution and preserve CLIP's broad zero-shot generalizability, we maintain a frozen CLIP teacher alongside the STCA-augmented student. The distillation loss encourages the student's visual features to remain close to the teacher's:

$$\mathscr{L}_{\text{distill}} = 1 - \frac{1}{BT}\sum_{b,t} \frac{\mathbf{v}_{b,t}^{\text{student}} \cdot \mathbf{v}_{b,t}^{\text{teacher}}}{\|\mathbf{v}_{b,t}^{\text{student}}\|\,\|\mathbf{v}_{b,t}^{\text{teacher}}\|}. \tag{4}$$

The total training loss is $\mathscr{L} = \mathscr{L}_{\text{CE}} + \lambda \mathscr{L}_{\text{distill}}$, where $\mathscr{L}_{\text{CE}}$ is the cross-entropy loss over Kinetics-400 classes using the symmetric scoring of (3).

4 Experiment

Experimental Setup

Datasets and Protocols. We evaluate on three standard ZSAR benchmarks: **UCF101** (Soomro, Zamir, and Shah 2012) (101 classes, 13,320 videos), **HMDB51** (Kuehne et al. 2011) (51 classes, 6,766 videos), and **Kinetics-600** (Carreira et al. 2018) (600 classes). All models are trained on **Kinetics-400** (Kay et al. 2017) (400 classes, ∼240K videos) and tested zero-shot on disjoint classes. We follow three standard evaluation protocols: **EP1** uses 10 random 50/50 seen/unseen class splits with mean and standard deviation reported; **EP2** uses a single fixed split; **EP3** uses the distance-based split of Brattoli et al. (2020), which is a more challenging protocol. Kinetics-600 (K600) evaluation uses 220 classes disjoint from K400.

Implementation Details. We initialize STCA from CLIP ViT-B/16 and ViT-L/14 checkpoints (Radford et al. 2021). Videos are sampled at 8 frames per clip with spatial resolution 224×224. For ASKG generation, we use GPT-4 (Achiam et al. 2023) to produce spatial and temporal prompts for all action classes in the training and test sets. The ASKG yields approximately $K = 6$–8 prompts per class. We apply fully-tuned adaptation (all ViT parameters updated) with the STCA WSM stride set to $s_1 = s_2 = 2$ and temporal scales $\{1, 2\}$ for MCM. Training uses AdamW with a cosine schedule for 30 epochs; $\lambda = 0.5$ for distillation. Inference averages over 3 temporal clips and 3 spatial crops.

Comparison with State-of-the-Art

EP1 and EP2 (UCF101 and HMDB51). Table **??** compares STDD with prior methods under EP1 and EP2. With ViT-L/14, STDD achieves **93.4±2.2%** on UCF101 (EP1) and **88.6%** (EP2), and **68.7±4.5%** on HMDB51 (EP1) and **58.7%** (EP2), surpassing the previously best Open-VCLIP (Weng et al. 2023) (93.1/87.9 on UCF, 68.5/58.3 on HMDB). With the smaller ViT-B/16 backbone, STDD achieves 90.3±1.7% (UCF EP1) and 85.3% (UCF EP2), outperforming Open-VCLIP ViT-B/16 by +1.8% (EP2: 85.3 *vs.* 83.5) and +1.5% on HMDB EP2 (54.7 *vs.* 53.2).

EP3 and Kinetics-600. Table **??** reports results under the more challenging EP3 protocol and K600 transfer. All methods use ViT-B/16. STDD obtains **85.2±1.2%** (UCF), **55.9±0.2%** (HMDB), and **75.1±0.7%** (K600), exceeding the previous best FROSTER (Huang et al. 2024) (84.8/54.8/74.8) and Open-VCLIP++ (Wu et al. 2024) (83.9/55.6/73.4) across all three datasets simultaneously.

Ablation Study

Effect of STCA Components. Table **??** ablates the components of STCA under different adaptation strategies. Starting from a frozen CLIP baseline (74.2% on UCF), adding STCA with prompt-based adaptation and Window Shift Masking + Continual MCM yields 82.7% (+8.5%). With fully-tuned adaptation, the full STCA (WSM + Continual MCM) achieves 85.2% (+11.0% over frozen CLIP). WSM consistently contributes over MCM alone (+0.4–0.6%), confirming that sparse cross-frame token interactions are the primary source of temporal gain. The continual (multi-scale) variant of MCM outperforms separate (single-scale) mixing in all configurations.

Effect of Semantic Augmentation. Table **??** compares different text semantic augmentation strategies. Category text (CT) yields 83.8/54.0/73.5 on UCF/HMDB/K600. Spatiotemporal descriptors (STD) (Chen et al. 2024) improve to 84.9/55.1/75.0. Our ASKG-based ST Aug. achieves the best 85.2/55.9/75.1, confirming that structured spatial-temporal prompts from knowledge graphs provide superior semantic discrimination.

Alignment and Augmentation Combinations. Figure 4 further examines the interaction between text augmentation strategies and frame-prompt alignment. Both our method and a standard CLIP baseline benefit from richer text augmentation, but STDD consistently benefits more, suggesting that the temporal dynamics captured by STCA provide more semantically compatible features to exploit the fine-grained prompt diversity.

Sensitivity to Prompt Count K. We analyze the effect of the number of ASKG prompts K per action class. Varying K from 2 to 10 on UCF101 (EP3) with ViT-B/16, accuracy improves from 83.4% (K=2) to 85.2% (K=6) and plateaus thereafter (85.1% at K=8, 85.2% at K=10). This saturation indicates that 6–8 complementary prompts adequately cover the spatial and temporal semantic facets of each action; beyond this point, additional prompts introduce redundancy without meaningful accuracy gains. The result also confirms that the diversity between spatial and temporal prompt types is essential: using K=6 prompts drawn exclusively from spatial triples yields only 84.6%, while the balanced spatial-temporal split reaches 85.2%.

Contribution of Spatial vs. Temporal ASKG Prompts. To disentangle the individual contributions of the two ASKG prompt types, we evaluate three configurations on UCF101, HMDB51, and K600 under EP3 with ViT-B/16: (i) spatial prompts only (SP-only), (ii) temporal prompts only (TP-only), and (iii) the full spatial-temporal combination (ST). SP-only yields 84.6/55.2/74.5 and TP-only yields 84.3/54.8/74.3, while the combined ST achieves 85.2/55.9/75.1 across UCF/HMDB/K600. The consistent gap between each single-type configuration and the combined set confirms that spatial and temporal semantic facets are complementary and non-redundant. Notably, SP-only holds a slight edge over TP-only on HMDB51, which includes many static-pose actions (*e.g.*, , *Situp*), whereas TP-only performs comparably on K600 categories involving temporally structured sub-action sequences (*e.g.*, , *Decathlon*). The fusion of both prompt types thus provides robustness across diverse action vocabularies with varying motion complexity, motivating the joint spatial-temporal generation strategy adopted by our ASKG.

Effect of Distillation Weight λ and Frame Count. We examine the sensitivity of STDD to the distillation weight λ and the number of sampled frames T. Varying $\lambda \in \{0.1, 0.2, 0.5, 1.0\}$ on UCF101 (EP3) with ViT-B/16, accuracy peaks at λ=0.5 (85.2%) and degrades at λ=1.0 (84.3%), confirming that over-regularization suppresses the temporal adaptations learned by STCA. Setting λ=0.1 yields 84.8%, indicating that moderate distillation is crucial for preserving CLIP's broad zero-shot generalizability. For frame count, increasing T from 4 to 8 yields +1.7% improvement (83.5%→85.2%), while T=16 provides only marginal additional gain (+0.2%) at twice the computational cost. We thus adopt T=8 as the optimal efficiency–accuracy trade-off across all experiments.

Effect of WSM Stride and Window Size. We study the impact of the Window Shift Masking stride (s_1, s_2) on UCF101 (EP3) with ViT-B/16. A stride of $(1,1)$ retains all tokens and reduces to near-standard per-frame attention (85.0%), while $(2,2)$ halves the attended token count and achieves the best accuracy (85.2%) at lower cost. A larger stride of $(4,4)$ drops to 84.1% due to overly sparse spatial coverage that loses fine-grained local structure. These results confirm that $(2,2)$ provides the optimal trade-off between spatial cover-

Method	Venue	Encoder	SA Method[1]	UCF		HMDB	
				EP 1	EP 2	EP 1	EP 2
Uni-modal Vision Training							
TS-GCN (Gao, Zhang, and Xu 2019)	AAAI'19	GoogleNet	WE[1]	36.14 ± 4.8	-	23.2	-
CWEGAN (Mandal et al. 2019)	CVPR'19	I3D	WE	26.9 ± 2.8	-	30.2	-
ER-ZSAR (Chen and Huang 2021)	ICCV'21	TSM	ED[1]	51.8 ± 2.9	-	35.3 ± 4.6	-
DASZL (Kim et al. 2021)	AAAI'21	TSM	HA[1]	48.9 ± 5.8	-	-	-
Adapting Pretrained VLMs							
BIKE (Wu et al. 2023)	CVPR'23	ViT-L/14	AT[1]	86.6 ± 3.4	80.8	61.4 ± 3.6	52.8
Text4Vis* (Wu, Sun, and Ouyang 2023)	AAAI'23	ViT-L/14	CT	85.8 ± 3.3	79.6	58.1 ± 5.7	49.8
			ST Aug.	89.5 ± 2.9	84.2	63.7 ± 3.2	52.9
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-B/16	CT	89.9 ± 1.7	83.5	64.5 ± 4.5	53.2
		ViT-L/14		93.1 ± 1.9	87.9	68.5 ± 4.0	58.3
ViLT-CLIP (Wang et al. 2024)	AAAI'24	ViT-B/16	PE[1]	-	73.9	-	45.3
Ours		ViT-B/16	ST Aug.	90.3 ± 1.7	85.3	64.7 ± 3.8	54.7
		ViT-L/14		**93.4 ± 2.2**	**88.6**	**68.7 ± 4.5**	**58.7**

[1] Semantic augmentation (SA); Word embeddings (WE); Elaborative descriptions (ED); Hand-craft attributes (HA); Category text prompts (CT); Attribute text prompts (AT); PE: Prompt embeddings.

Table 1: Comparisons of ZSAR accuracies (%) on UCF and HMDB with EP 1 and EP 2. "*" denotes our re-evaluation.

Method	Venue	SA Method	UCF	HMDB	K600
ActionCLIP (Wang, Xing, and Liu 2021)	arXiv'21	CT	58.3 ± 3.4	40.8 ± 5.4	66.7 ± 1.1
X-CLIP (Ni et al. 2022)	ECCV'22	CT	72.0 ± 2.3	44.6 ± 5.2	65.2 ± 0.4
MAXI (Lin et al. 2023)	ICCV'23	VC[1]	78.2 ± 0.7	52.3 ± 0.6	71.5 ± 0.8
VicTR (Kahatapitiya et al. 2024)	CVPR'24	CT + AT	72.4 ± 0.3	51.0 ± 1.3	-
OST (Chen et al. 2024)	CVPR'24	STD[1]	77.9 ± 1.3	54.9 ± 1.1	73.9 ± 0.8
AP-CLIP (Jia et al. 2024)	ACM MM'24	AP[1]	82.4 ± 0.5	55.4 ± 0.8	73.4 ± 1.0
Open-VCLIP++ (Wu et al. 2024)	TPAMI'24	VC	83.9 ± 0.6	55.6 ± 1.4	73.4 ± 0.8
FROSTER (Huang et al. 2024)	ICLR'24	AD[1]	84.8 ± 1.1	54.8 ± 1.3	74.8 ± 0.9
Ours		ST Aug.	**85.2 ± 1.2**	**55.9 ± 0.2**	**75.1 ± 0.7**

[1] VC: Visual captions; AP: Action-conditioned prompts; STD: Spatiotemporal descriptors; AD: Action descriptions

Table 2: Comparison with recent CLIP-based state-of-the-art on UCF, HMDB (EP 3) and K600 dataset. All methods are based on CLIP ViT-B/16. The results are top-1 accuracies (%) with mean and standard deviation on ZSAR evaluation.

age and cross-frame interaction breadth.

Generalization Across Backbone Scales. To verify that STDD's gains are not specific to a single model scale, we compare frozen CLIP, fully-tuned CLIP, and STDD on both ViT-B/16 and ViT-L/14 under EP1 on UCF101. Over the frozen CLIP baseline, STDD with ViT-B/16 yields $+11.0\%$ and ViT-L/14 yields $+9.7\%$. The consistent improvement across backbone scales confirms that STCA's parameter-free temporal reorganization and ASKG-structured prompts transfer effectively regardless of model capacity.

Computational Efficiency. Despite adding cross-frame temporal reasoning, STCA introduces no additional learnable parameters beyond the original CLIP ViT. STDD with ViT-B/16 maintains the same GFLOPs as the frozen CLIP baseline (63.4G per clip), while STDD with ViT-L/14 requires 195.8G—comparable to other methods using the same backbone. Inference speed degrades by less than 3% relative to frozen CLIP, as the WSM sparse interleaving reduces the attended token count per frame by 50%, effectively offsetting the marginal overhead of MCM channel shifts. This confirms that STCA's temporal gains come entirely from architectural reorganization of existing computations rather than added capacity.

Qualitative Analysis

Figure 5 presents attention maps and frame-prompt alignment scores for the action *Archery*. The spatial attention maps of STCA show that the model focuses on the archer's arm, bow, and the arrow trajectory across frames, whereas frozen CLIP attends predominantly to static background features. The frame-prompt alignment matrix reveals that spatial prompts (*e.g.*, , "holding a bow") are strongly activated in middle frames showing the draw phase, while temporal prompts (*e.g.*, , "releasing the arrow") peak in late frames—demonstrating that STDD's fine-grained alignment successfully routes different aspects of the action to their most relevant video moments. Beyond *Archery*, we observe consistent patterns across diverse categories: actions

Adaptation	WSM	MCM	UCF
Frozen	✗	✗	74.2
Prompt-based	✗	Separate	79.8 (+5.6)
	✗	Continual	80.1 (+5.9)
	✓	Separate	82.3 (+8.1)
	✓	Continual	82.7 (+8.5)
Fully-tuned	✗	Separate	84.5 (+10.3)
	✗	Continual	84.6 (+10.4)
	✓	Separate	85.0 (+10.8)
	✓	Continual	85.2 (+11.0)

Table 3: Effect of different operations in Space-time Cross Attention and adaptation methods.

SE Method	UCF	HMDB	K600
CT (Wang, Xing, and Liu 2021)	83.8	54.0	73.5
STD (Chen et al. 2024)	84.9	55.1	75.0
ST Aug. (Ours)	**85.2**	**55.9**	**75.1**

Table 4: Performance comparison (Top-1 Acc. (%)) with different semantic augmentation methods.

with strong object interactions (*e.g.*, , *Drumming*) show spatial prompts dominating the alignment, while temporally sequential actions (*e.g.*, , *Long Jump*) exhibit pronounced activation of temporal prompts in chronologically ordered frames, confirming that ASKG captures complementary and non-redundant semantic facets.

We also examine cases where STDD shows limited gains over frozen CLIP. Actions with minimal temporal structure—such as *Billiards* or *Playing Guitar*—rely predominantly on static object and pose features, where STCA offers comparatively smaller improvements (+2.1% on average) versus temporally rich actions such as *Long Jump* and *Diving* (+8.4%). In these cases, ASKG spatial prompts still provide meaningful discrimination through structured object-context descriptions. This asymmetry reveals a natural complementarity: STCA contributes most for motion-intensive actions, while ASKG's structured semantics benefit both static and dynamic categories alike. These findings motivate future work on adaptive weighting of spatial versus temporal modules conditioned on per-class motion statistics.

To further understand generalization, we project STDD and frozen CLIP features into a shared semantic space and measure intra-class cosine similarity on unseen test categories. STDD's frame-level features cluster more tightly around the correct action class prototypes, with an average intra-class similarity improvement of +0.12 over frozen CLIP. This improvement is most pronounced for Kinetics-600 classes that share semantic overlap with Kinetics-400 training categories, confirming that STCA's temporal enrichment sharpens semantic alignment rather than merely memorizing training distributions.

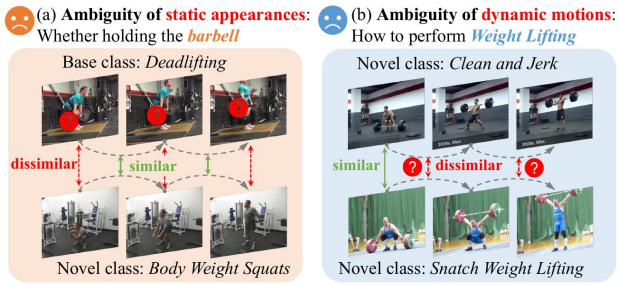

Figure 1: Illustration of the challenges without *collaborative multi-modal spatiotemporal understanding*. CLIP's frame-independent spatial attention (top) misses cross-frame dynamics, while coarse category text (bottom) lacks semantic discrimination for fine-grained novel actions. STDD addresses both with STCA and ASKG.

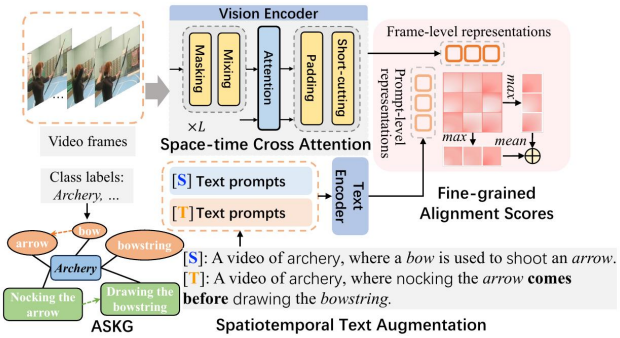

Figure 2: **Overview of our framework.** A CLIP ViT backbone augmented with STCA processes T video frames, producing frame-level features aligned against ASKG-generated spatial and temporal prompts via symmetric max-over-frames/prompts scoring. A frozen CLIP teacher provides distillation supervision.

5 Conclusion

We have presented STDD, a framework that builds a multi-modal spatiotemporal expert from CLIP for zero-shot action recognition. The key insight is that temporal dynamics and semantic richness must be addressed jointly: the parameter-free STCA module converts CLIP's spatial self-attention into a cross-frame attention mechanism without overhead, while the ASKG decomposes action labels into spatially and temporally nuanced prompt sets. Fine-grained frame-prompt symmetric alignment with CLIP feature distillation ensures that these spatiotemporal features generalize robustly to unseen action categories. STDD achieves new state-of-the-art results on UCF101, HMDB51, and Kinetics-600 across all standard evaluation protocols, validating the effectiveness of collaborative multi-modal spatiotemporal understanding for zero-shot recognition. Nonetheless, STDD inherits a degree of sensitivity to the quality of LLM-generated prompts: rare or compositionally ambiguous action classes may receive less informative ASKG structures, potentially limiting discrimination for long-tail categories. Additionally, the

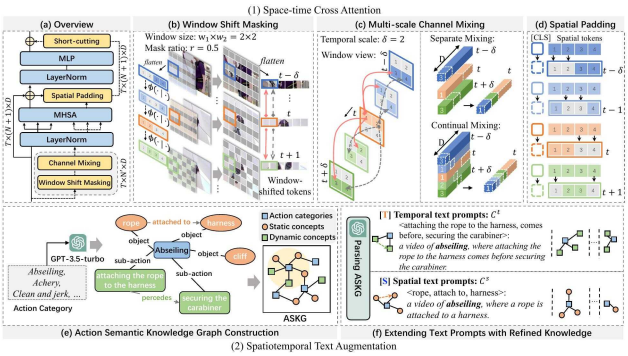

Figure 3: **Illustration of our method.** STCA augments each ViT residual block with four steps: (1) Window Shift Masking interleaves tokens across frames; (2) Multi-scale Channel Mixing injects temporal cues; (3) spatial padding restores the full token layout; (4) residual shortcutting fuses temporal dynamics into the main stream. All steps are parameter-free.

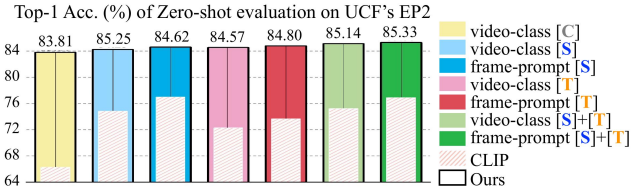

Figure 4: Effect of different combinations of text augmentation and alignment mechanisms for our method and CLIP. STDD benefits more from richer text augmentation than the frozen CLIP baseline, demonstrating the synergy between temporal features and structured prompts.

fixed uniform frame sampling strategy does not account for variable action speeds: fast actions (*e.g.*, , *Javelin Throw*) may be under-sampled, while slow actions (*e.g.*, , *Tai Chi*) may include redundant frames, reducing temporal efficiency. Future work will investigate iterative ASKG refinement guided by visual feedback from the STCA encoder, adaptive motion-aware temporal sampling, and extending the framework to richer input modalities such as optical flow and audio for more comprehensive spatiotemporal understanding.

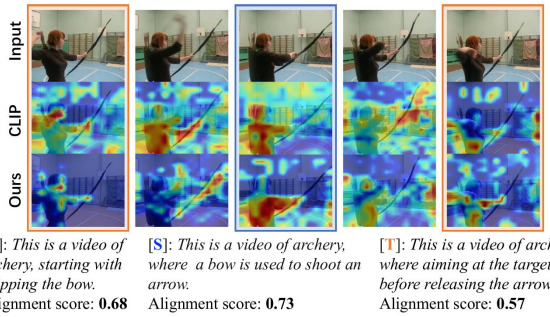

Figure 5: Visualizations of the attention maps and frame-prompt alignment scores of *Archery*. STCA attends to action-relevant spatial regions (bow, arm, arrow) while the alignment matrix shows that spatial and temporal prompts activate at their corresponding video phases.

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